5.13.4 Constructing a parallelogram when its one side, one diagonal and one angle are given

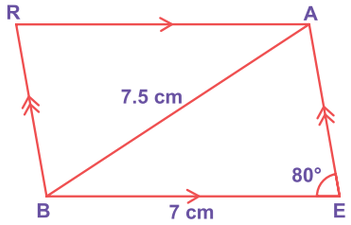
Example 5.33

Construct a parallelogram BEAR with BE=7*cm*, BA=7.5*cm* and $\angle BEA = 80°$. Also find its area.

Solution:

Given:

BE=7*cm*, BA=7.5*cm* and $\angle BEA = 80°$



Rough diagram

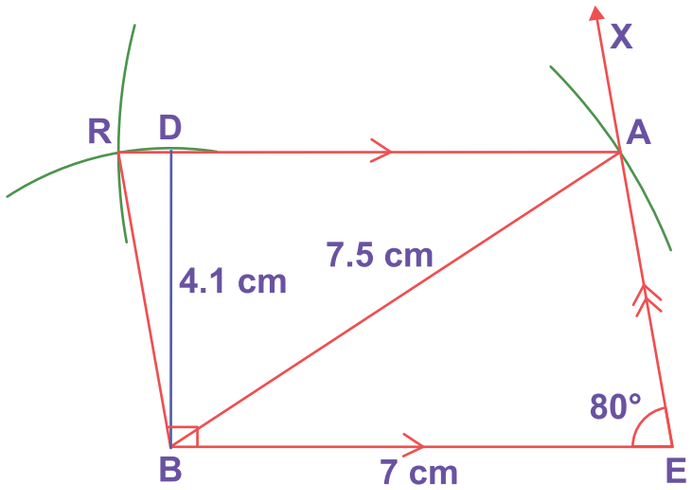


Fig. 5.52

Steps:

1. Draw a line segment BE=7*cm*.
2. Make an angle $\angle BEX = 80°$ at E on $\overline{BE}$.
3. With B as centre, draw an arc of radius 7.5*cm* cutting EX at A and Join BA.
4. With B as centre, draw an arc of radius equal to the length of $\overline{AE}$.
5. With A as centre, draw an arc of radius 7*cm*. Let both arcs cut at R.
6. Join BR and AR.
7. BEAR is the required parallelogram.

Calculation of Area:

Area of the parallelogram BEAR = *bh* sq.units

= 7×4.1= 28.7 *sq.cm*

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